Samagalla Sukesh Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.70
Sri Chaitanya Junior Kalasala	Class XII (TSBIE)	2023	96.80%
Narayana High School	Class X (SSC)	2021	100.00%

WORK EXPERIENCE

Media.net | Software Engineering Intern

- **Built an event-driven automation service integrating YouTrack, GitLab, and LLM agents** to automatically maintain task context, process code changes, and respond to user requests without manual task maintenance.
- **Engineered a per-task knowledge base in PostgreSQL** that continuously aggregates requirements, engineering decisions, MR changes, and task activity, enabling agents to retrieve persistent context for downstream reasoning.
- **Designed an agentic comment workflow** that classifies user intent, gathers relevant task/MR/PRD context, and executes targeted actions including description updates, Markdown summaries, or contextual comments.
- **Improved pipeline efficiency and reliability** by replacing MR-triggered LLM rewrites with deterministic processing, implementing idempotent updates and bot self-echo protection, eliminating unnecessary LLM calls and redundant YouTrack API requests.

PROJECTS

RMGA: Robust Motion Gated Adaptation for Video Streams | Computer Vision

- Developed RMGA, a Test-Time Adaptation framework for 3D CNNs that mitigates environmental domain shifts, improving **UCF50 test accuracy by up to 15.81%**.
- Engineered a memory-efficient adaptation pipeline using forward-only BatchNorm warmup, entropy-based uncertainty estimation, and spatio-temporal motion masking to selectively adapt dynamic regions.
- Designed a motion-gated entropy + consistency loss objective, achieving **13.57% higher accuracy than ViTTA** on localized-motion classes while reducing inference overhead for edge deployment.

Financial Decision Making | Natural Language Processing

- Developed a Unified GRU-based sequential trading model integrating NLP and market signals to generate daily BUY/HOLD/SELL predictions for BTC and TSLA in the CLEF 2026 FinMMEval Lab.
- Integrated FinBERT sentiment embeddings with technical market features and designed a custom asymmetric loss function that penalizes directional errors according to their potential trading impact.
- Implemented strict temporal data-splitting and anti-data-leakage protocols, achieving **19.91% cumulative return** and a **2.840 Sharpe ratio** on unseen test data.

Multi-Agent Wildfire Search and Adaptive Survivor Rescue Coordination | Multi-Agent Systems (Ongoing)

- Developing a **decentralized multi-agent system** for cooperative UAV search, survivor tracking, task allocation, and rescue coordination in a dynamic wildfire environment.
- Implementing a **fire propagation and risk model** incorporating terrain, vegetation, trails, and wind conditions to generate dynamic wildfire scenarios for UAV coordination.
- Integrating **heterogeneous UAV agents, dynamic communication, RL-based task allocation, and safety constraints** to evaluate coordination under UAV failures, communication disruptions, and changing fire conditions.

RISC-V Simulator | Computer Architecture

- Built a C-based RISC-V instruction-set simulator supporting instruction execution, register operations, and memory access management.
- Engineered **Direct-Mapped, Fully Associative, and Set-Associative cache architectures** to model cache lookup and memory access behavior.
- Implemented FIFO, LRU, LFU, and RANDOM replacement policies to evaluate cache hit/miss rates and replacement behavior across memory access patterns.

SKILLS

- **Programming Language:** Python, C++, C, RISC-V assembly.
- **Database Management:** PostgreSQL, MySQL.
- **Technical Skills:** LangChain, LangGraph, FastAPI, PyTorch, MCP, Agentic AI, Automation.
- **Tools:** Docker, Postman, Git, GitHub, Jupyter Notebook, Google Colab.

RELEVANT COURSES

- **Computer Science:** Database Management Systems, Computer Architecture, Data Structures, Algorithms, SWE.
- **Artificial Intelligence:** Computer Vision, Natural Language Processing, AI in Finance, Multi-Agent Systems, Data Mining, Deep Learning, Reinforcement Learning, Fundamentals of Machine Learning.
- **Mathematics:** Matrix Theory, Bayesian Statistics, Applied Statistics, Probability and Random Variables, Discrete Mathematics, Calculus.

EXTRACURRICULAR

- **NSO Hockey:** I am part of the IIT Hyderabad's Hockey Team, and an active participant in the Diesta Competitions.
- In my leisure time, I play cricket, badminton, and Volleyball. I am a **Sports Enthusiast.**

POSITIONS OF RESPONSIBILITY

Team Manager, Valorant Competition | IIT Hyderabad **Esports** Community

- Led and managed a team of peers in the institute-level Valorant tournament, securing **1st place** among competing teams.